

Generation Interconnection System Impact Study Report

For Queue Project Q18-0989

Greenbrae 138 kV
November 01, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-0989. The Interconnection Customer proposes a 206.9 MW solar photovoltaic facility interconnecting at the Greenbrae 138 kV substation in the Virginia area. The request was received on May 08, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 355 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 19 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$91,626,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Fresh Meadows Substation - Lakeshore 138 kV Line
Contingency: Hinman - Floyd Substation 138 kV
Loading: 138.0% (Rating: 172 MVA)
DFAX: 100.0%
Pre-project loading: 18.0%
2. Facility: Pinnacle Peak Srp - East Portal 138 kV
Contingency: Hinman - Floyd 138 kV L
Loading: 127.1% (Rating: 232 MVA)
DFAX: 100.0%
Pre-project loading: 37.8%
3. Facility: E Portal - Crystal Mountain Generation 138 kV
Contingency: Hinman - Floyd 138 kV L
Loading: 126.8% (Rating: 224 MVA)
DFAX: 100.0%
Pre-project loading: 34.4%
4. Facility: Fall River - Crystal Mtn Generation 138 kV Line
Contingency: Hinman - Floyd 138 kV L
Loading: 125.8% (Rating: 219 MVA)
DFAX: 100.0%
Pre-project loading: 31.2%
5. Facility: Team - Fresh Meadows 138 kV line
Contingency: Floyd - Hinman 138 kV
Loading: 125.4% (Rating: 207 MVA)
DFAX: 100.0%
Pre-project loading: 25.4%
6. Facility: Team - Fall Riv 138 kV
Contingency: Floyd - Hinman 138 kV
Loading: 124.8% (Rating: 212 MVA)

DFAX: 100.0%

Pre-project loading: 27.3%

7. Facility: Hinman Substation - Floyd 138 kV line

Contingency: Pinnacle Peak Srp - E Portal 138 kV

Loading: 122.2% (Rating: 241 MVA)

DFAX: 100.0%

Pre-project loading: 36.3%

8. Facility: Kirksville - Floyd 138 kV

Contingency: Pinnacle Peak Srp - E Portal 138 kV

Loading: 120.0% (Rating: 238 MVA)

DFAX: 100.0%

Pre-project loading: 33.2%

9. Facility: Kirksville - Oteen 138 kV

Contingency: Pinnacle Peak Srp - E Portal 138 kV

Loading: 119.3% (Rating: 232 MVA)

DFAX: 100.0%

Pre-project loading: 30.1%

10. Facility: Highway 59 - Lakeshore 138 kV line

Contingency: Floyd - Hinman 138 kV

Loading: 118.2% (Rating: 188 MVA)

DFAX: 100.0%

Pre-project loading: 8.2%

11. Facility: Shullsburg - Willard Johnson 138 kV line

Contingency: Pinnacle Peak Srp - East Portal 138 kV L

Loading: 118.1% (Rating: 222 MVA)

DFAX: 100.0%

Pre-project loading: 24.7%

12. Facility: Oteen Substation - Willard Johnson 138 kV

Contingency: Pinnacle Peak Srp - E Portal 138 kV

Loading: 117.7% (Rating: 228 MVA)

DFAX: 100.0%

Pre-project loading: 27.0%

13. Facility: Midway - South Christiansburg 138 kV

Contingency: Pinnacle Peak Srp - East Portal Substation 138 kV

Loading: 111.9% (Rating: 218 MVA)

DFAX: 62.2%

Pre-project loading: 53.1%

14. Facility: Happy Canyon - S Christiansburg 138 kV

Contingency: East Portal Substation - Pinnacle Peak Srp 138 kV

Loading: 109.1% (Rating: 247 MVA)

DFAX: 62.2%

Pre-project loading: 57.0%

15. Facility: Happy Canyon - Edgemont 138 kV L

Contingency: Pinnacle Peak Srp Substation - E Portal 138 kV

Loading: 107.0% (Rating: 273 MVA)

DFAX: 62.2%

Pre-project loading: 59.9%

16. Facility: W Cookeville - Greenbrae 138 kV line

Contingency: Clinch Riv - Carlto Hills 345 kV

Loading: 106.0% (Rating: 237 MVA)

DFAX: 100.0%

Pre-project loading: 18.7%

17. Facility: Edgemont Substation - Shullsburg 138 kV
Contingency: Pinnacle Peak Srp - E Portal 138 kV line
Loading: 105.3% (Rating: 300 MVA)
DFAX: 62.2%
Pre-project loading: 62.4%

18. Facility: W Cookeville Substation - Riley 138 kV
Contingency: Clinch Riv - Carlton Hills 345 kV
Loading: 103.6% (Rating: 226 MVA)
DFAX: 100.0%
Pre-project loading: 11.9%

19. Facility: Riley - Hinman 138 kV
Contingency: Eufaula - Beaver 500 kV
Loading: 102.5% (Rating: 221 MVA)
DFAX: 100.0%
Pre-project loading: 9.0%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Fresh Meadows Substation - Lakeshore 138 kV Line	Rebuild line with higher-capacity conductor	\$10,232,000	100% (\$10,232,000)
East Portal Substation - Pinnacle Pea Srp 138 kV Line	Rebuild line with higher-capacity conductor	\$5,140,000	100% (\$5,140,000)
E Portal - Crystal Mountain Generation 138 kV	Rebuild line with higher-capacity conductor	\$2,903,000	100% (\$2,903,000)
Crystal Mtn Generation - Fall River 138 kV L	Rebuild line with higher-capacity conductor	\$5,453,000	100% (\$5,453,000)
Team - Fresh Meadows 138 kV line	Rebuild line with higher-capacity conductor	\$8,094,000	100% (\$8,094,000)
Team - Fall Riv 138 kV	Rebuild line with higher-capacity conductor	\$3,945,000	100% (\$3,945,000)
Hinman - Floyd 138 kV	Rebuild line with higher-capacity conductor	\$7,615,000	100% (\$7,615,000)
Kirksville - Floyd 138 kV	Rebuild line with higher-capacity conductor	\$8,145,000	100% (\$8,145,000)
Oteen - Kirksville 138 kV	Rebuild line with higher-capacity conductor	\$8,395,000	100% (\$8,395,000)
Highway 59 - Lakeshore 138 kV line	Rebuild line with higher-capacity conductor	\$6,849,000	100% (\$6,849,000)
Willard Johnson - Shullsburg 138 kV	Rebuild line with higher-capacity conductor	\$4,653,000	100% (\$4,653,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Oteen Substation - Willard Johnson 138 kV	Rebuild line with higher-capacity conductor	\$4,772,000	100% (\$4,772,000)
Midway - South Christiansburg 138 kV	Rebuild line with higher-capacity conductor	\$4,247,000	79% (\$3,342,000)
Hapy Canyon - S Christiansburg 138 kV	Reconductor line	\$1,083,000	79% (\$852,000)
Edgemont - Happy Canyon 138 kV line	Reconductor line	\$752,000	79% (\$592,000)
West Cookeville - Greenbrae 138 kV Line	Reconductor line	\$2,922,000	100% (\$2,922,000)
Shullsburg - Edgemont 138 kV	Reconductor line	\$1,227,000	79% (\$965,000)
West Cookeville - Riley 138 kV	Reconductor line	\$4,508,000	100% (\$4,508,000)
Riley - Hinman 138 kV line	Reconductor line	\$2,249,000	100% (\$2,249,000)
	Total allocated to project		\$91,626,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.